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1. Draw a histogram showing distribution of heights across species for penguin dataset

Create a stack plot and step plot- – Write down your interpretations

```
In [15]: import seaborn as sns  
import matplotlib.pyplot as plt
```

```
In [16]: sns.get_dataset_names()
```

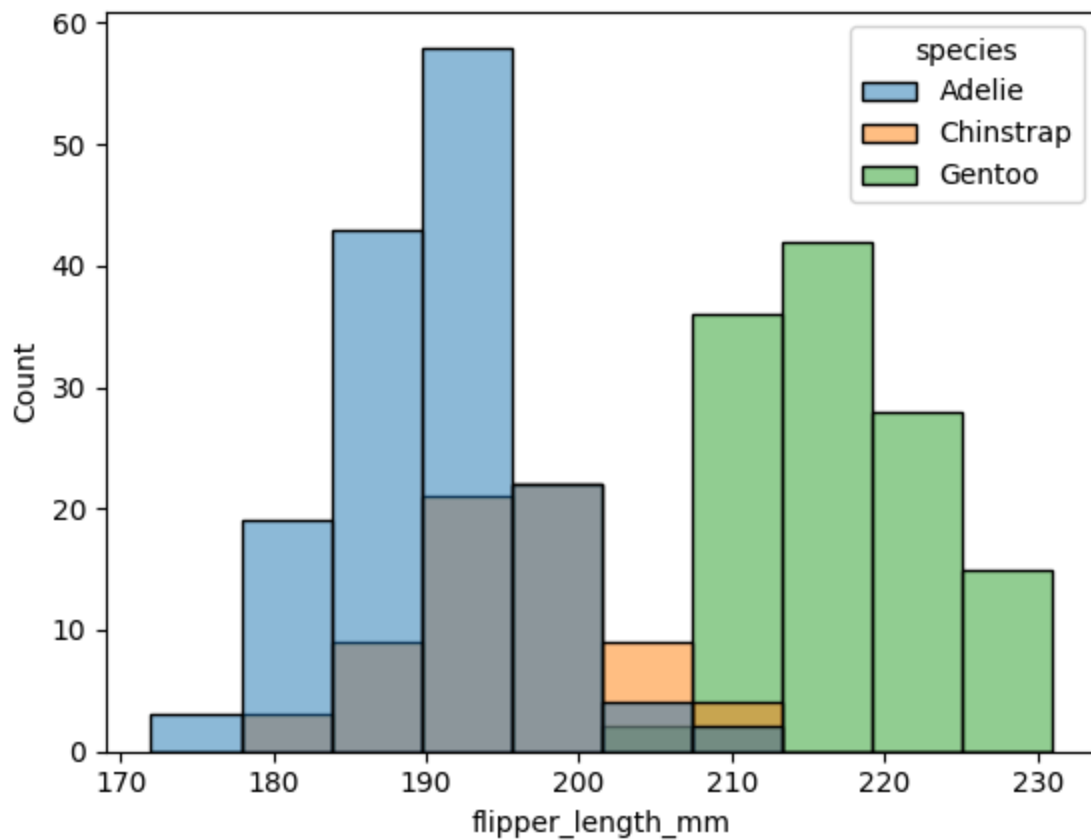
```
Out[16]: ['anagrams',  
          'anscombe',  
          'attention',  
          'brain_networks',  
          'car_crashes',  
          'diamonds',  
          'dots',  
          'dowjones',  
          'exercise',  
          'flights',  
          'fmri',  
          'geyser',  
          'glue',  
          'healthexp',  
          'iris',  
          'mpg',  
          'penguins',  
          'planets',  
          'seaice',  
          'taxis',  
          'tips',  
          'titanic']
```

```
In [17]: df = sns.load_dataset('penguins')  
df
```

Out[17]:

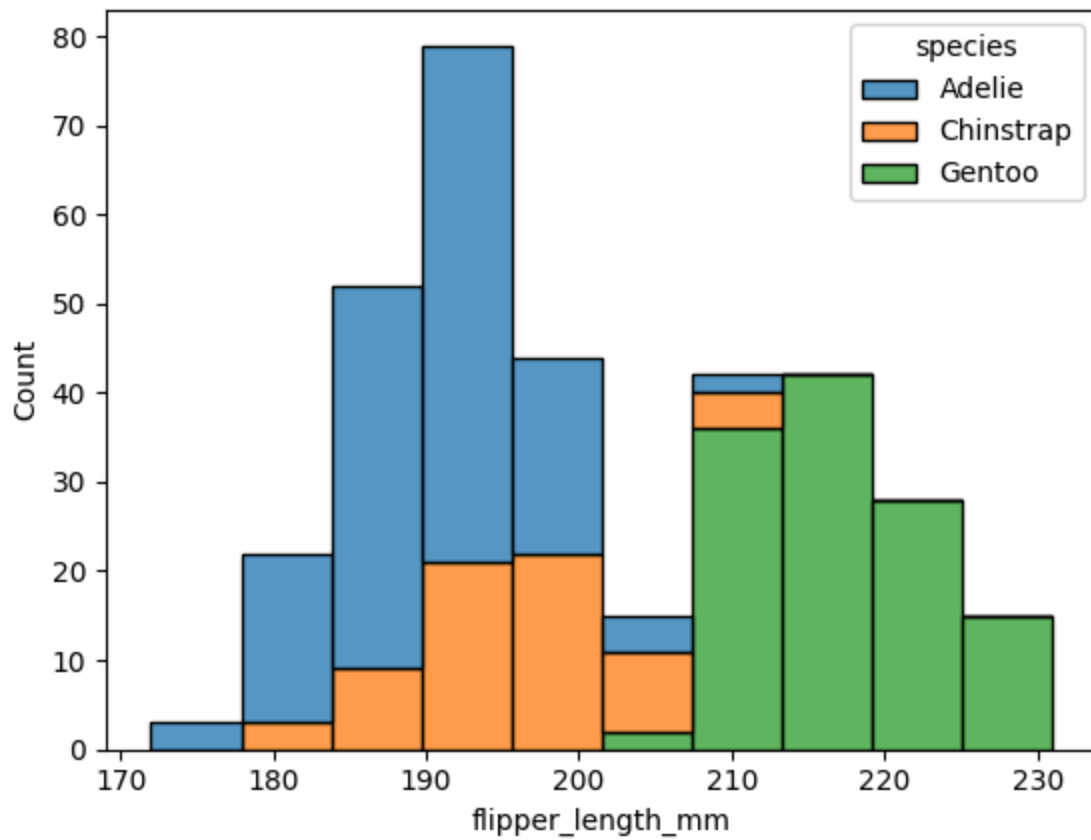
	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0
3	Adelie	Torgersen	NaN	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0
...	...	...	...	...	...	...
339	Gentoo	Biscoe	NaN	NaN	NaN	NaN
340	Gentoo	Biscoe	46.8	14.3	215.0	4850.0
341	Gentoo	Biscoe	50.4	15.7	222.0	5750.0
342	Gentoo	Biscoe	45.2	14.8	212.0	5200.0
343	Gentoo	Biscoe	49.9	16.1	213.0	5400.0

344 rows × 7 columns

In [18]: `sns.histplot(df, x = 'flipper_length_mm', hue='species')`Out[18]: `<Axes: xlabel='flipper_length_mm', ylabel='Count'>`

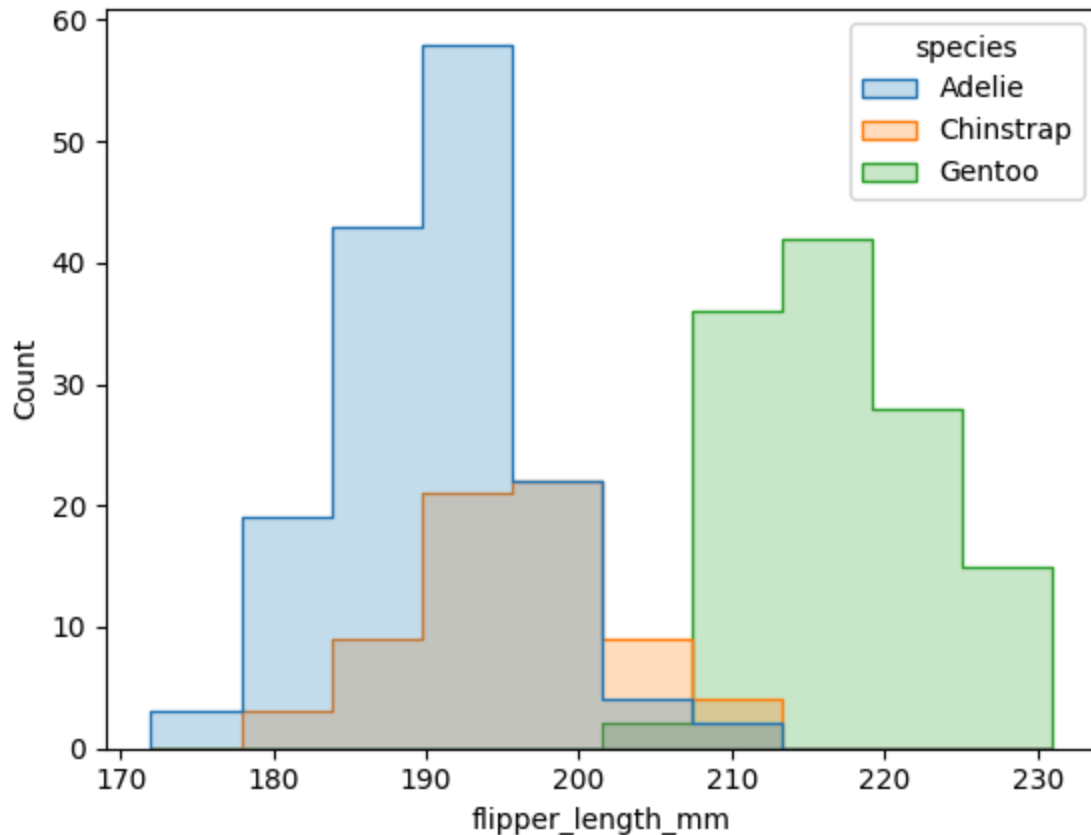
```
In [19]: sns.histplot(df, x = 'flipper_length_mm', multiple='stack', hue = 'species')
```

```
Out[19]: <Axes: xlabel='flipper_length_mm', ylabel='Count'>
```



```
In [20]: sns.histplot(df, x = 'flipper_length_mm', hue = 'species', element='step')
```

```
Out[20]: <Axes: xlabel='flipper_length_mm', ylabel='Count'>
```



From the histogram plots, we can observe that:

- Gentoo penguins consistently show the longest flipper lengths among all species
- Chinstrap penguins have intermediate flipper lengths
- Adelie penguins have the shortest flipper lengths
- The stacked plot shows clear separation between species' size distributions
- The step plot highlights the distinct ranges of flipper lengths for each species

Q2. Draw a bivariate KDE plot to show relationship between horsepower and displacement [use mpg dataset] – Write down your interpretations

```
In [21]: mpg = sns.load_dataset('mpg')
mpg
```

Out[21]:

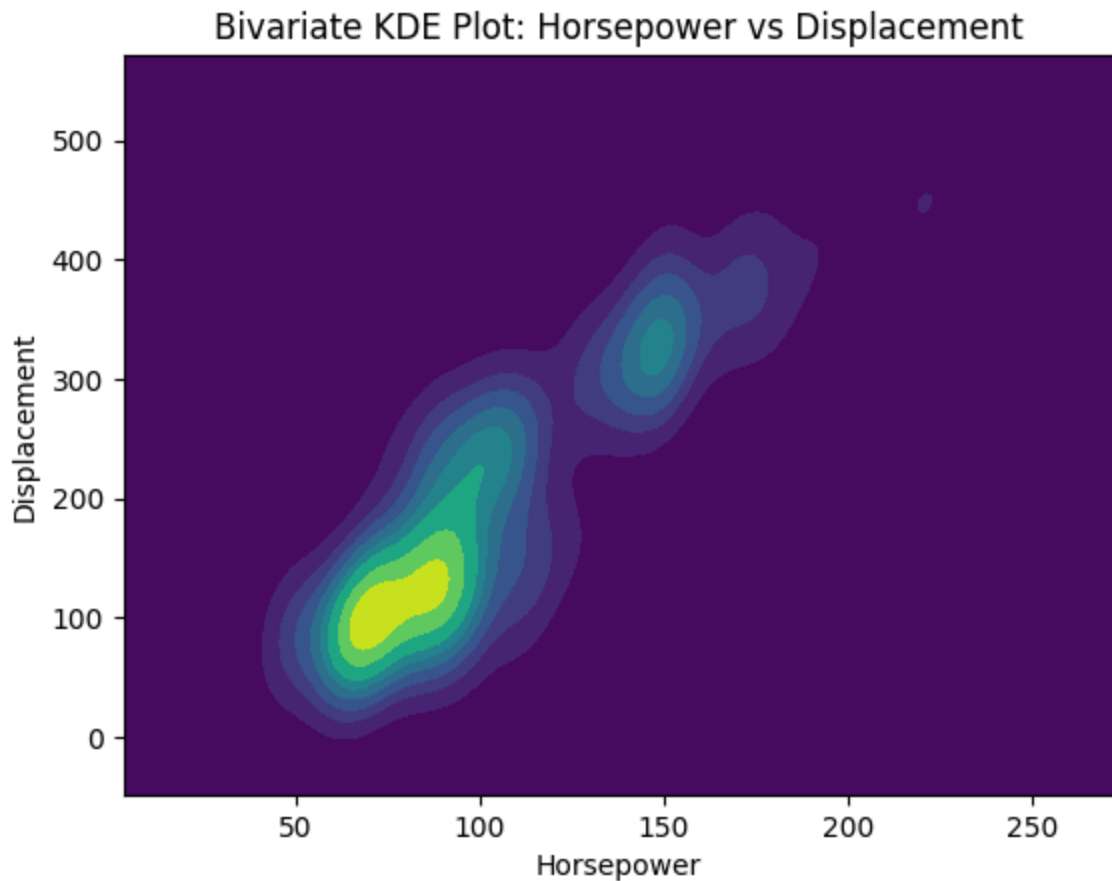
	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin
0	18.0	8	307.0	130.0	3504	12.0	70	usa
1	15.0	8	350.0	165.0	3693	11.5	70	usa
2	18.0	8	318.0	150.0	3436	11.0	70	usa
3	16.0	8	304.0	150.0	3433	12.0	70	usa
4	17.0	8	302.0	140.0	3449	10.5	70	usa
...	...	...	...	...	...	...	...	...
393	27.0	4	140.0	86.0	2790	15.6	82	usa
394	44.0	4	97.0	52.0	2130	24.6	82	europe
395	32.0	4	135.0	84.0	2295	11.6	82	usa
396	28.0	4	120.0	79.0	2625	18.6	82	usa
397	31.0	4	119.0	82.0	2720	19.4	82	usa

398 rows × 9 columns



```
In [29]: # Create a bivariate KDE plot
sns.kdeplot(data=mpg, x='horsepower', y='displacement', cmap='viridis', fill=True,1
plt.title('Bivariate KDE Plot: Horsepower vs Displacement')
plt.xlabel('Horsepower')
plt.ylabel('Displacement')
```

```
Out[29]: Text(0, 0.5, 'Displacement')
```



- If there's a positive relationship, you'll see the darker colors running from bottom-left to top-right
 - Higher density (darker areas) shows where most cars cluster
 - The contour lines (like topographic maps) show regions of equal density
3. Draw a correlation heatmap for diamonds dataset, add annotation, change colour pallet and write your interpretation

```
In [24]: diamonds = sns.load_dataset('diamonds')  
diamonds
```

Out[24]:

	carat	cut	color	clarity	depth	table	price	x	y	z
0	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	3.98	2.43
1	0.21	Premium	E	SI1	59.8	61.0	326	3.89	3.84	2.31
2	0.23	Good	E	VS1	56.9	65.0	327	4.05	4.07	2.31
3	0.29	Premium	I	VS2	62.4	58.0	334	4.20	4.23	2.63
4	0.31	Good	J	SI2	63.3	58.0	335	4.34	4.35	2.75
...	...	...	...	...	...	...	...	...	...	...
53935	0.72	Ideal	D	SI1	60.8	57.0	2757	5.75	5.76	3.50
53936	0.72	Good	D	SI1	63.1	55.0	2757	5.69	5.75	3.61
53937	0.70	Very Good	D	SI1	62.8	60.0	2757	5.66	5.68	3.56
53938	0.86	Premium	H	SI2	61.0	58.0	2757	6.15	6.12	3.74
53939	0.75	Ideal	D	SI2	62.2	55.0	2757	5.83	5.87	3.64

53940 rows × 10 columns

```
In [25]: diamonds = diamonds.loc[:, 'depth':]
diamonds
```

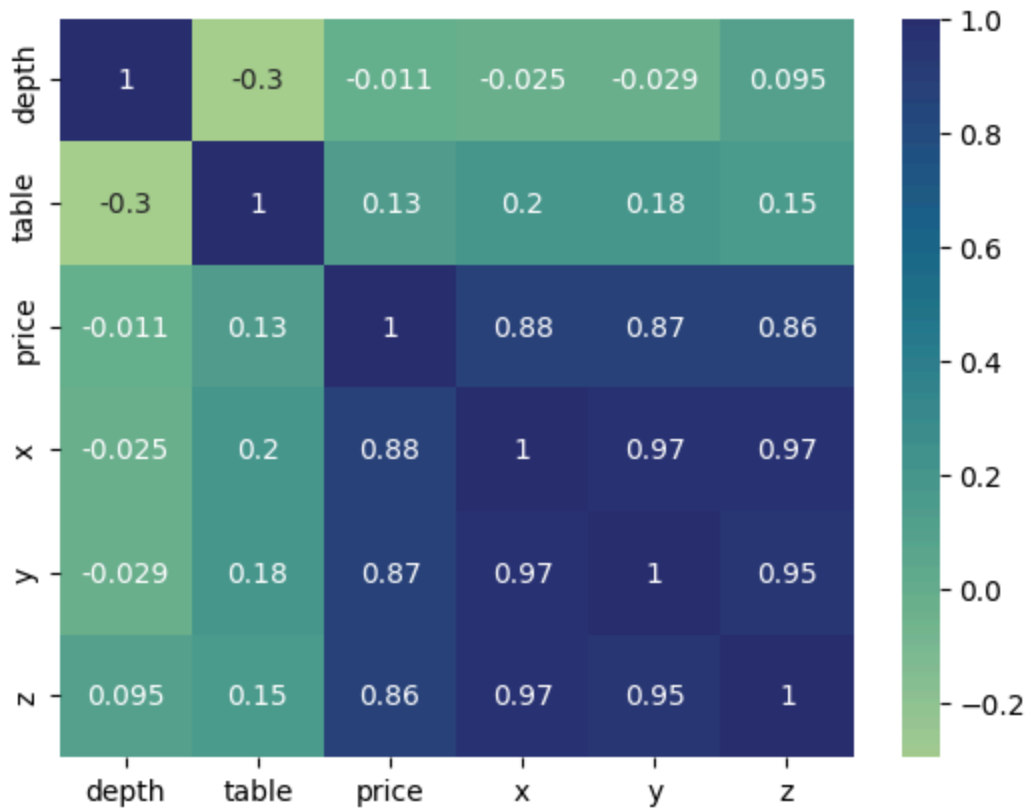
Out[25]:

	depth	table	price	x	y	z
0	61.5	55.0	326	3.95	3.98	2.43
1	59.8	61.0	326	3.89	3.84	2.31
2	56.9	65.0	327	4.05	4.07	2.31
3	62.4	58.0	334	4.20	4.23	2.63
4	63.3	58.0	335	4.34	4.35	2.75
...	...	...	...	...	...	...
53935	60.8	57.0	2757	5.75	5.76	3.50
53936	63.1	55.0	2757	5.69	5.75	3.61
53937	62.8	60.0	2757	5.66	5.68	3.56
53938	61.0	58.0	2757	6.15	6.12	3.74
53939	62.2	55.0	2757	5.83	5.87	3.64

53940 rows × 6 columns

```
In [26]: sns.heatmap(diamonds.corr(), annot=True, cmap='crest' )
```

Out[26]: <Axes: >



Diamond Correlation Heatmap Interpretation:

- There is a strong positive correlation between carat and price
- As diamond size (carat) increases, the price tends to increase proportionally
- The x, y, z dimensions show strong correlations with each other
- Darker blue regions indicate stronger positive correlations
- Lighter blue regions indicate weaker correlations
- The diagonal shows perfect correlation (1.0) as expected

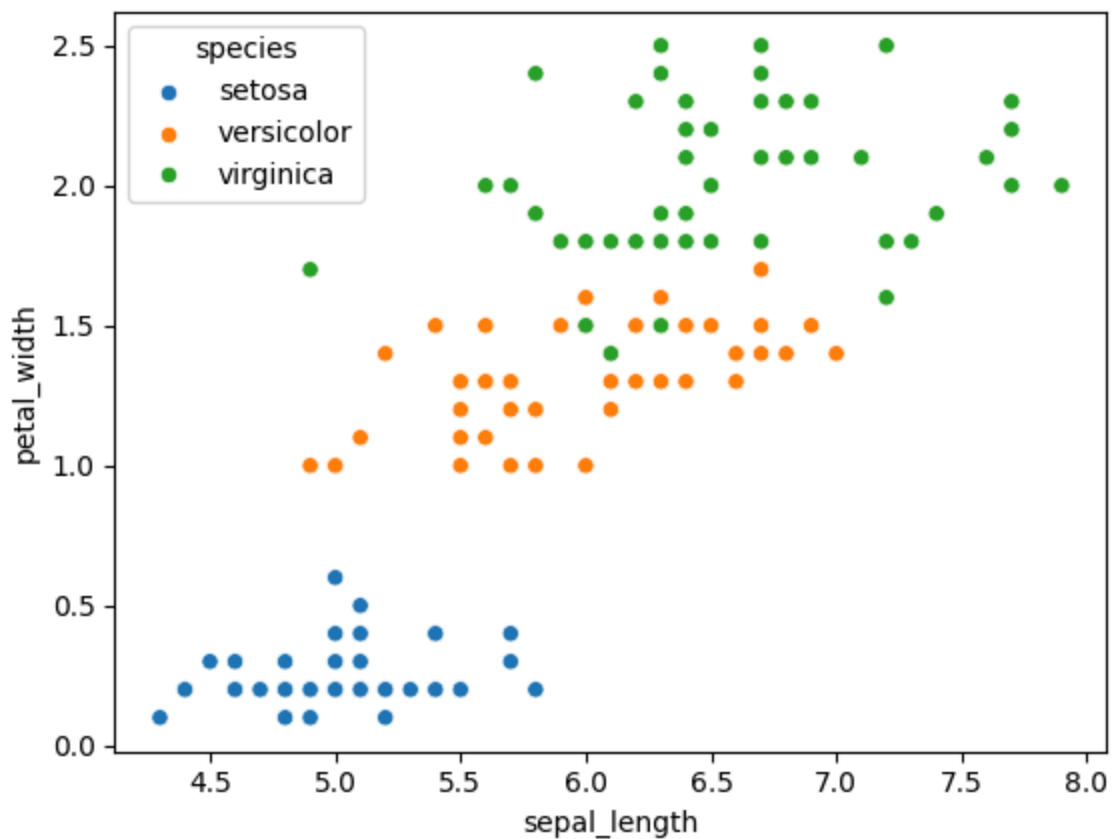
4. Draw a scatter plot to visualizing relationships between flower measurements. [e.g. sepal_length, petal_width] across the species. Write your interpretations

```
In [27]: iris = sns.load_dataset('iris')
iris
```


Out[27]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [28]: `sns.scatterplot(iris, x='sepal_length', y='petal_width', hue = 'species')`Out[28]: `<Axes: xlabel='sepal_length', ylabel='petal_width'>`

- Setosa flowers have small petal widths and sepal lengths, forming a distinct cluster.
- Versicolor and Virginica have larger measurements and slightly overlap.
- There is a positive correlation: as sepal length increases, petal width increases.
- The scatter plot shows that species can be visually distinguished based on these measurements.